

QUAID-I-AZAM UNIVERSITY
DEPARTMENT OF COMPUTER SCIENCE (BS 5th)

Terminal Examination Fall, 2020

Course No. ST-101 Course Title: Probability and Statistics

Max. Marks: 50

Time Allowed: 3 Hours

Teacher's Name: Mr. Abdullah Iqbal

Date: 15-JAN-2021

Note: Please attempt five questions. All questions carry equal marks.

Q.1 (a) What methods would employ in collection of statistical data when field of inquiry is (i) small, (ii) fairly large and (iii) very large, if you are to pay due regard to accuracy, labour, cost and time? **(04)**

(b) The marks obtained by 32 students in Statistics and Mathematics are given below. **(06)**

Prepare a two way table.

Statistics	15	9	16	17	15	7	5	13
Mathematics	15	15	14	13	9	3	3	16
Statistics	8	11	1	9	10	11	10	0
Mathematics	3	17	5	11	13	14	10	2
Statistics	2	11	4	14	14	2	19	6
Mathematics	18	7	1	17	12	8	19	10
Statistics	13	12	6	15	8	2	10	18
Mathematics	11	8	4	18	11	8	13	18

Q.2 (a) Find the probability that the sum of the scores on three rolls of an ordinary fair die is less than 5. **(04)**

(b) Every Saturday a man invites his sister to the theatre or to the cinema. 70 % of his invitation are to the theatre and 90% of these are accepted. His sister rejects 40% of his invitation to the cinema. Find the probability that the brother's invitation is accepted on any particular Saturday. **(03)**

(c) After settlement, average daily wages in factory had increased from Rs. 120 to Rs. 180 and the S.D had increased from Rs. 2.5 to Rs. 3.75. After settlement the wages have become higher and more uniform. Comment. **(03)**

Q.3 (a) Explain clearly the concepts of Random Variable. What is a distribution function? **(03)**

(b) The following table shows the joint probability distribution of random variables X and Y **(04)**

X	1	2
Y		
1	2/15	1/15
3	4/15	1/15
5	3/15	4/15

Compute the conditional distribution $f(x/y)$. Check whether X and Y are independent random variables.

(c) For what value of A, the function defined as below will be a p.d.f **(03)**

$$f(x) = Ax^3(1-x), \quad 0 \leq x \leq 1$$

$$= 0 \quad \text{elsewhere}$$

Also find its mean and variance.

Q.4 (a) Describe a Poisson Distribution with the help of at least two real life examples? **(02)**

(b) According to national fire protection report more than one-third of all hotel fires are caused by guest room furnishing. Suppose that guest room furnishings caused a hotel fire is 35%. What is the probability that three of next four hotel fires will be caused by the guest room furnishings, assuming independence of events? **(04)**

(c) Given that the height of college boys is normally distributed with mean 5'-2" and standard deviation 4", and that the minimum height required for joining the N.C.C. is 5'-4"; find the percentage of boys who would be rejected on account of their height. **(04)**

Q.5 (a) How could you define Normal Distribution in relation with its parameters? **(02)**

(b) An irregular six-faced die is thrown and the expectation that in 10 throws it will give five even numbers is twice the expectation that it will give four even numbers. How many times, 10,000 sets of 10 throws would you expect it to give no even number? **(04)**

(c) A random sampling of 4 members club has shown that 3 prefer no smoking in the clubhouse dining room. What is the probability that this will occur if in fact only 20% of members prefer no smoking in the dining room. **(04)**

BEST OF LUCK